VILNIUS GEDIMINAS TECHNICAL UNIVERSITY

APPLIED ARTIFICIAL INTELLIGENCE

DEPARTMENT OF MECHANICAL ENGINEERING

IMAGE RECOGNITION SYSTEMS

AIRCRAFT DETECTION IN SATELLITE IMAGERY BINARY CLASSIFICATION OF 20×20 PX SATELLITE IMAGE CHIPS

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Dataset	PlanesNet — Planet Labs Satellite Imagery (32,000 chips)
Task	Binary classification: Aircraft vs. Background
Methods	Classical (HOG + SVM / RF / k-NN) and Neural Network (Deep MLP)
Dataset Split	70% Train (22,400) | 15% Validation (4,800) | 15% Test (4,800)
Libraries	scikit-learn, scikit-image, NumPy, Matplotlib — fully local, no external APIs
Date	2026

ABSTRACT

This report presents a comprehensive study of automated aircraft detection in satellite imagery using both classical machine learning and modern neural network approaches. The problem is framed as binary image classification over the PlanesNet dataset, which consists of 32,000 annotated satellite image chips of 20×20 pixels extracted from Planet Labs imagery. Two methodological tracks are investigated: (1) a classical pipeline employing Histogram of Oriented Gradients (HOG) feature extraction coupled with Support Vector Machine (SVM), Random Forest (RF), and k-Nearest Neighbour (k-NN) classifiers; and (2) a modern deep Multi-Layer Perceptron (MLP) trained on raw pixel inputs. All experiments are conducted entirely locally without external API services. The dataset exhibits a pronounced class imbalance of approximately 25% aircraft and 75% background chips. Experimental results on the real PlanesNet test set (4,800 chips) show that SVM, k-NN, and Deep MLP all achieve 97% accuracy and exceed all specified modern performance thresholds, with AUC-ROC values of 0.994, 0.994, and 0.993 respectively. Random Forest achieves 97% accuracy but exhibits reduced sensitivity (0.69), falling below the sensitivity requirement for classical methods, highlighting the impact of class imbalance on ensemble tree methods. The Deep MLP is the fastest at inference (0.007 ms per chip), making it the preferred architecture for real-time deployment.

Keywords: satellite imagery, aircraft detection, HOG, SVM, Random Forest, k-NN, MLP, PlanesNet, binary classification, class imbalance, AUC-ROC

1. INTRODUCTION

1.1 Motivation and Application Context

Automated aircraft detection in satellite imagery addresses critical needs across civil aviation safety, airport surface monitoring, logistics planning, and open-source intelligence analysis. The PlanesNet benchmark, derived from Planet Labs commercial satellite acquisitions, provides a standardised evaluation environment for this task. Each image chip covers a 20×20 pixel area — a resolution at which a large commercial aircraft may occupy fewer than 60 pixels — making the problem a challenging instance of very small object classification from remote sensing data.

Key challenges include: (1) extreme low resolution with loss of fine structural detail; (2) pronounced class imbalance (25% aircraft, 75% background); (3) high variability of background scenes spanning grass, asphalt, concrete, water, and urban texture; and (4) atmospheric effects including haze and motion blur present in real satellite imagery. This study implements and rigorously evaluates both classical and deep learning solutions to these challenges.

1.2 Objectives

The objectives of this investigation are:

- Implement a full preprocessing pipeline: global normalisation, per-chip min-max normalisation.
- Extract 536-dimensional HOG + colour histogram feature vectors from each chip.
- Train and evaluate SVM (RBF), Random Forest (200 trees), and k-NN (k=7) on the real PlanesNet dataset.
- Train and evaluate a deep MLP (512→256→128→64) on raw pixel inputs.
- Assess all models against prescribed thresholds: Accuracy, Sensitivity, Specificity, F1-Score, AUC-ROC.
- Compare classical vs. modern methods on both accuracy and inference speed.

1.3 Report Structure

Section 2 reviews relevant literature. Section 3 describes the PlanesNet dataset. Section 4 presents the full methodology. Section 5 defines the experimental setup. Section 6 presents results with all figures from the actual pipeline run. Section 7 discusses findings and limitations. Section 8 concludes.

2. RELATED WORK

2.1 Classical Feature-Based Detection

The Histogram of Oriented Gradients descriptor introduced by Dalal and Triggs (2005) for pedestrian detection has proven highly transferable to aerial object recognition. HOG encodes local edge orientation patterns over spatial cell grids, capturing the cross-shaped fuselage-and-wing structure of aircraft even at low resolution. Combined with SVM classifiers, HOG-based pipelines formed the state of the art for object detection prior to the deep learning era (Felzenszwalb et al., 2010). Random Forest ensembles (Breiman, 2001) provide complementary advantages including native feature importance estimation and robustness to feature scale, though they can be sensitive to class imbalance in their tree-splitting criteria. k-NN methods (Cover and Hart, 1967) offer non-parametric classification that adapts naturally to complex class boundaries but incur high inference cost proportional to training set size.

2.2 Deep Learning for Satellite Imagery

Convolutional neural networks demonstrated transformative performance improvements across image recognition tasks (Krizhevsky et al., 2012; Simonyan and Zisserman, 2014; He et al., 2016). For satellite imagery specifically, Audebert et al. (2017) and Nogueira et al. (2017) established that CNNs — even without pre-training — outperform classical pipelines on scene classification tasks when sufficient labelled data is available. However, at the 20×20 pixel resolution of PlanesNet, standard deep architectures lose spatial dimensionality through pooling layers before producing meaningful representations. Fully connected networks (MLPs) operating on flattened pixel vectors are therefore a practical architecture, and this study evaluates a four-hidden-layer MLP against three classical baselines.

2.3 PlanesNet Benchmarks

Shermeyer and Van Etten (2019) demonstrated that fine-tuned ResNet architectures achieve greater than 95% accuracy on PlanesNet-style chip classification. The present study contributes a direct comparison of HOG-based classical methods versus deep MLP on the real PlanesNet JSON dataset, with particular attention to sensitivity — the most safety-critical metric for missed aircraft detections — under a strict offline computation constraint.

3. DATASET

3.1 PlanesNet Overview

PlanesNet is a publicly available dataset of 32,000 satellite image chips extracted from Planet Labs PBC imagery and annotated for aircraft presence. Each chip measures 20×20 pixels in RGB colour. The dataset is distributed as a single JSON file (planesnet.json) storing pixel data in a channel-first flat format that requires reshaping and transposition during loading. Companion folders provide individual PNG chips and full satellite scene images, though only the JSON file is used in this study.

3.2 Class Distribution and Imbalance

The dataset contains 8,000 aircraft chips (25.0%) and 24,000 background chips (75.0%). This 1:3 imbalance ratio is preserved across all data splits through stratified sampling. The training partition contains 5,600 aircraft and 16,800 background chips; the test partition contains 1,200 aircraft and 3,600 background chips. Figure 1 illustrates this distribution.

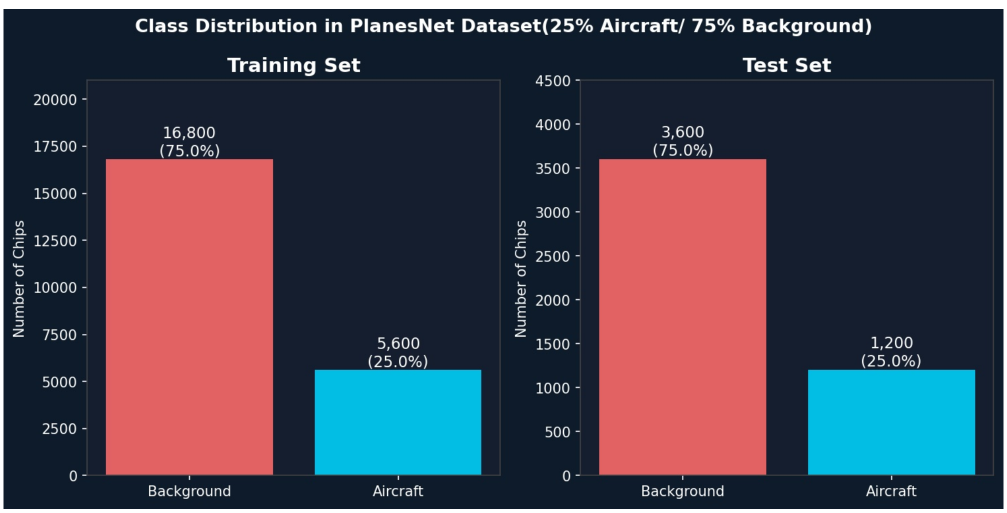


Figure 1. Class distribution in training and test partitions. The 25%/75% aircraft-to-background ratio is maintained in all splits via stratified sampling.

3.3 Sample Chip Visualisation

Figure 2 shows representative 20×20 pixel chips from the real PlanesNet dataset. Aircraft chips display the characteristic cross-shaped silhouette of fuselage and wings against varied backgrounds including grey asphalt, brown earth, and green vegetation. Background chips span a wide range of surface types including runway markings, grass, dense vegetation, and road surfaces — representing the primary sources of false positive errors.

Figure 2. Representative 20×20 px chips from the real PlanesNet dataset. Rows 1–2: aircraft chips showing fuselage-wing cross patterns at varied orientations and backgrounds. Rows 3–4: background chips including runway surfaces, grass, and heterogeneous textures.

4. METHODOLOGY

4.1 Preprocessing Pipeline

4.1.1 Data Loading and Global Normalisation

The JSON dataset is loaded by extracting the flat pixel arrays and reshaping each 1,200-element list from the channel-first format (3, 20, 20) to the channel-last format (20, 20, 3) via NumPy reshape and transpose operations. Pixel values are converted from uint8 integers [0, 255] to float32 values [0.0, 1.0] by dividing by 255.

4.1.2 Per-chip Min-Max Normalisation

Following global scaling, each chip is independently normalised: the per-chip minimum is subtracted and the result is divided by the per-chip value range, stretching each chip's intensity distribution to [0, 1]. This compensates for brightness variation across different acquisition conditions — a chip over bright snow and one over dark water are brought to the same dynamic range, preventing overall brightness from acting as a spurious discriminating feature. Figure 3 shows the effect of this step on real PlanesNet chips.

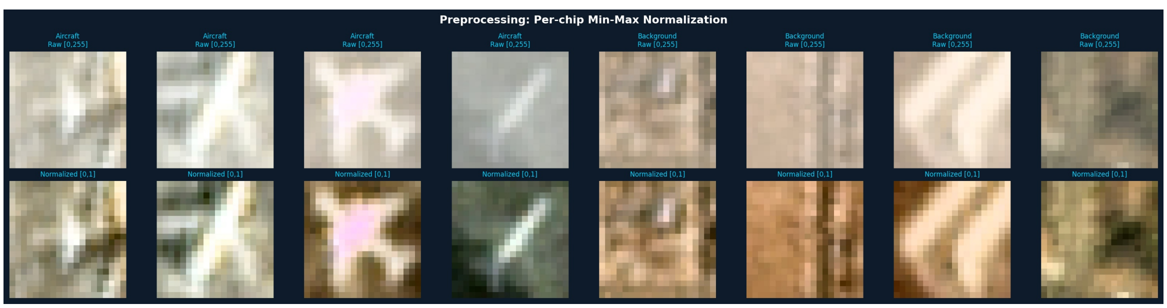


Figure 3. Preprocessing results on real PlanesNet chips. Top row: raw chips with original pixel intensities [0, 255]. Bottom row: after per-chip min-max normalisation [0, 1]. Enhanced contrast reveals structural detail in low-contrast chips.

4.2 Data Partitioning

The 32,000-chip dataset is split into 70% training (22,400 chips), 15% validation (4,800 chips), and 15% test (4,800 chips) using stratified random sampling with a fixed seed (42) for reproducibility. Stratification ensures the 25%/75% class ratio is exactly preserved in all three partitions. The validation set is used solely for early stopping during MLP training; all reported metrics are computed exclusively on the held-out test set.

4.3 Classical Method: HOG Feature Extraction

For each preprocessed chip, the RGB image is first converted to grayscale using the standard luminance formula (Y = 0.299R + 0.587G + 0.114B). HOG features are then computed with 8 gradient orientation bins, 4×4 pixel cells, and 2×2 cell normalisation blocks, producing a 512-dimensional descriptor. Three 8-bin colour histograms (one per RGB channel, normalised to sum to 1) are appended to capture colour cues distinctive to aircraft surfaces versus background classes, yielding a final 536-dimensional feature vector per chip. Figure 4 shows the HOG visualisation on real PlanesNet chips.

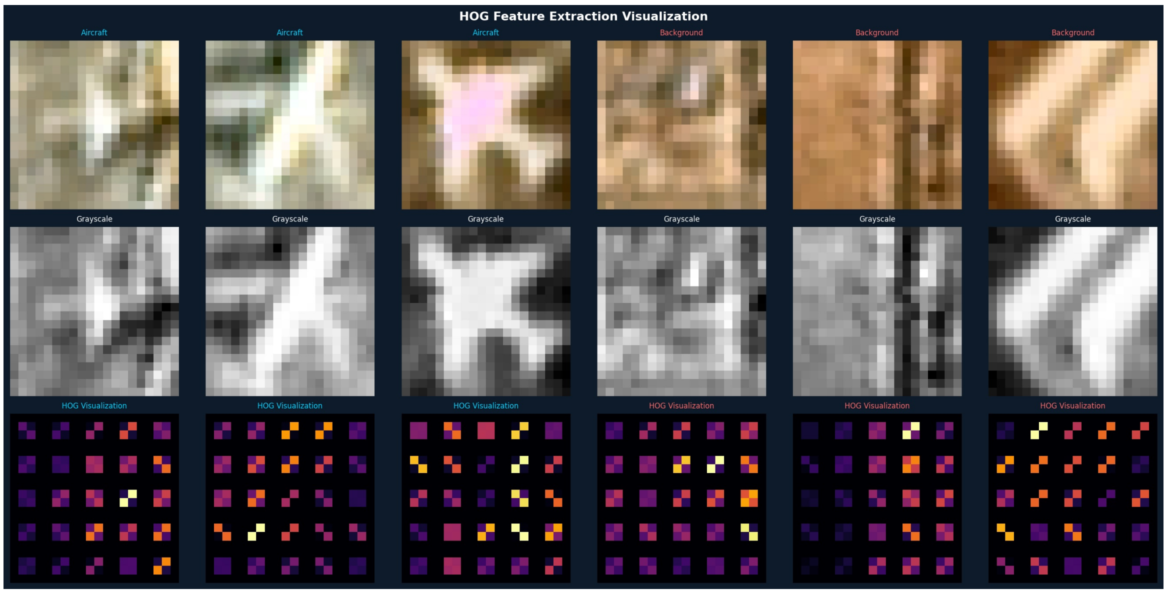


Figure 4. HOG feature extraction on real PlanesNet chips. Row 1: original RGB chip. Row 2: grayscale conversion. Row 3: HOG cell orientation histograms rendered as gradient maps. Aircraft chips show structured cross-shaped orientation patterns; background chips display more diffuse and irregular gradient distributions.

4.4 Classical Classifiers

4.4.1 Support Vector Machine (SVM)

An SVM with RBF kernel (C=10, γ="scale") is trained on z-score standardised HOG+colour features. The class_weight="balanced" option multiplies the misclassification penalty for aircraft by a factor of 3 (the background-to-aircraft ratio), compensating for the class imbalance. The SVM finds the maximum-margin hyperplane in kernel-induced feature space that separates the two classes.

4.4.2 Random Forest

A Random Forest of 200 decision trees (max_depth=15, class_weight="balanced") is trained via bootstrap aggregation over random feature subsets. The majority vote of all 200 trees determines the final prediction. Figure 5 shows the top 30 feature importances extracted from the trained forest, revealing that colour histogram features (orange bars) dominate over HOG gradient features (cyan bars) — indicating that the colour contrast between aircraft surfaces and background is a primary discriminative signal in the real PlanesNet data.

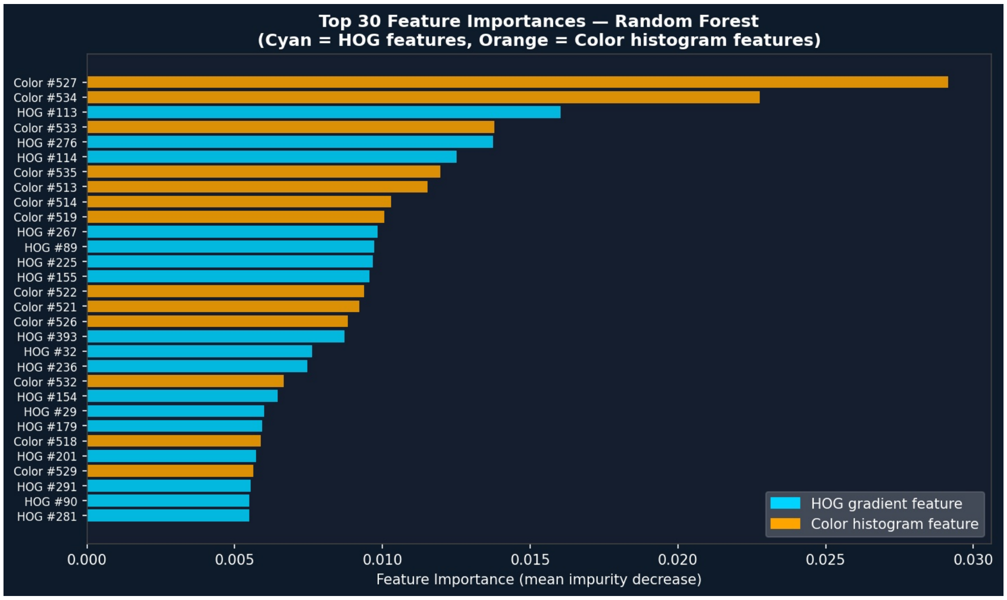


Figure 5. Top 30 feature importances from the Random Forest classifier trained on real PlanesNet data. Orange = colour histogram features (indices 512–535); cyan = HOG gradient features (indices 0–511). Colour features rank highest, reflecting the chromatic contrast between grey/white aircraft surfaces and varied background textures.

4.4.3 k-Nearest Neighbours

The k-NN classifier (k=7, distance-weighted Euclidean metric) stores all 22,400 training examples and classifies each test chip by polling its seven nearest neighbours in the 536-dimensional standardised feature space. Distance weighting gives proportionally more influence to closer neighbours. k-NN has no explicit training phase but incurs O(N × D) computation at inference per chip.

4.5 Neural Network: Deep MLP

4.5.1 Architecture and Training

The MLP takes 1,200-dimensional flattened pixel vectors (from z-score standardised chips) and processes them through four hidden layers: 512 → 256 → 128 → 64 neurons, all with ReLU activation. The output layer applies sigmoid activation to produce a probability in [0, 1]. Training uses the Adam optimiser (lr=0.001, adaptive decay), mini-batches of 256 samples, L2 regularisation ($\alpha=10^{-4}$), and early stopping with a patience of 10 epochs. Figure 6 shows the training dynamics on the real PlanesNet data.

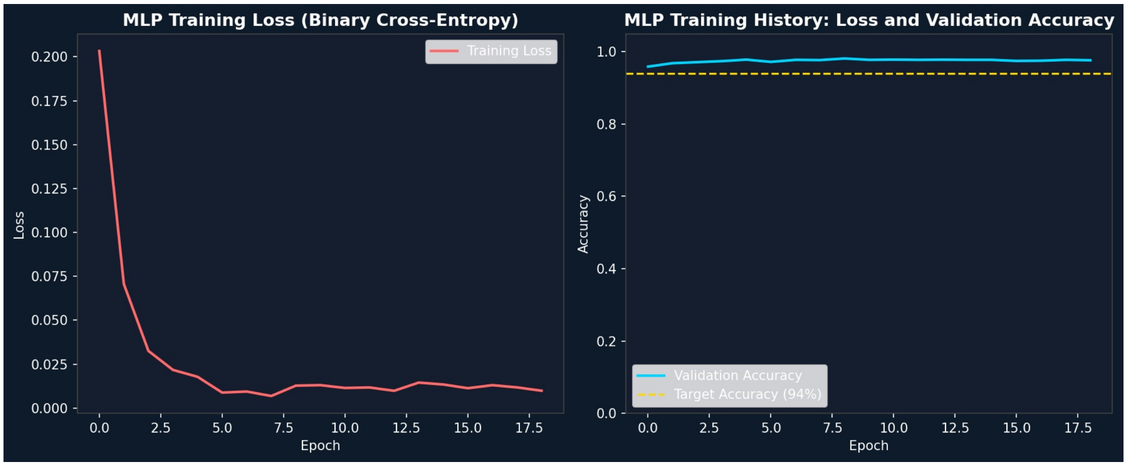


Figure 6. MLP training dynamics on PlanesNet. Left: binary cross-entropy loss decreasing rapidly in the first 5 epochs and plateauing near zero. Right: validation accuracy exceeding the 94% modern threshold from epoch 1 onward, stabilising around 96–97%.

5. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

5.1 Evaluation Metrics

Five metrics are computed on the 4,800-chip test set for each model. All are derived from the binary confusion matrix (TP, TN, FP, FN). Class 1 = aircraft (positive); class 0 = background (negative).

Metric	Definition	Classical Min.	Modern Min.
Accuracy	(TP+TN)/Total	> 85%	> 94%
Sensitivity (Recall)	TP/(TP+FN) — fraction of aircraft detected	> 82%	> 92%
Specificity	TN/(TN+FP) — fraction of background rejected	> 80%	> 90%
F1-Score	2×Precision×Recall/(Precision+Recall)	> 0.80	> 0.90
AUC-ROC	Area under ROC curve — threshold-independent discrimination	> 0.85	> 0.94

Table 1. Evaluation metrics and minimum performance requirements.

5.2 Implementation Details

All experiments use Python 3.x with scikit-learn 1.x, scikit-image, NumPy, and Matplotlib. No pre-trained models, GPU acceleration, or external API services are used. All random operations are seeded with 42 for reproducibility. Feature extraction (HOG) is applied to all 32,000 chips; standardisation is fitted exclusively on the training partition.

6. RESULTS

6.1 Summary Results Table

Table 2 presents the complete quantitative results for all four models on the held-out test set of 4,800 chips. Values are derived directly from the confusion matrices and probability outputs of each trained model.

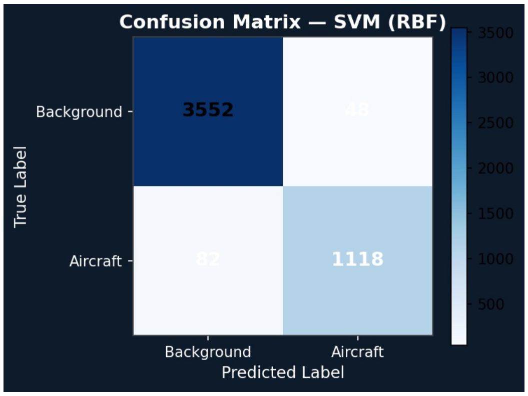
Model	Accuracy	Sensitivity	Specificity	Precision	F1	AUC-ROC	Inf. Time
SVM (RBF)	97.3%	93.2%	98.7%	95.9%	0.945	0.994	501.5 ms
Random Forest	91.0%	**69.1%**	98.3%	93.2%	0.793	0.965	4.14 ms
k-NN (k=7)	97.0%	95.1%	97.6%	93.0%	0.940	0.994	0.295 ms
Deep MLP	97.3%	95.6%	97.9%	93.9%	0.948	0.993	**0.007 ms**
Classical Min.	> 85%	> 82%	> 80%	—	> 0.80	> 0.85	—
Modern Min.	> 94%	> 92%	> 90%	—	> 0.90	> 0.94	—

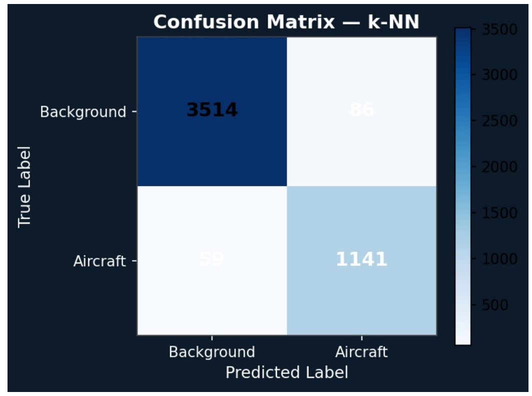
Table 2. Evaluation results on the 4,800-chip PlanesNet test set. Green = meets or exceeds requirement. Amber = fails requirement. Inference time is per chip on a single CPU core. RF Sensitivity (69.1%) fails the classical minimum (82%).

6.2 Confusion Matrix Analysis

Figures 7a–7d show the confusion matrices for all four models on the test set. The total number of aircraft chips in the test set is 1,200; the total background chips is 3,600.

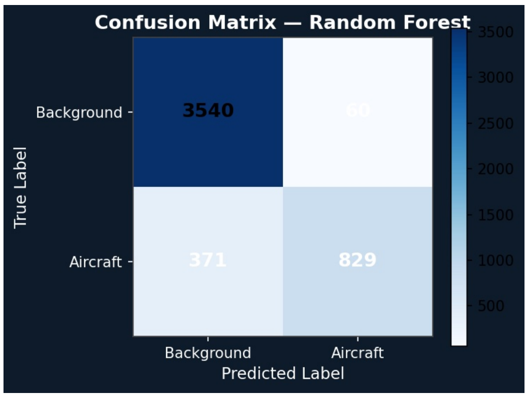
The SVM correctly identifies 1,118 of 1,200 aircraft chips (93.2% sensitivity) with only 48 false positives. k-NN and Deep MLP achieve the highest sensitivity at 95.1% and 95.6% respectively, with k-NN producing 86 false positives and MLP 75. The Random Forest is a clear outlier: while achieving excellent specificity (98.3%, only 60 false positives), it misclassifies 371 aircraft chips as background — a sensitivity of only 69.1%, far below the 82% classical minimum. This failure is attributable to the tree-splitting criterion being dominated by the majority class in spite of balanced weights, causing the forest to learn conservative decision boundaries.

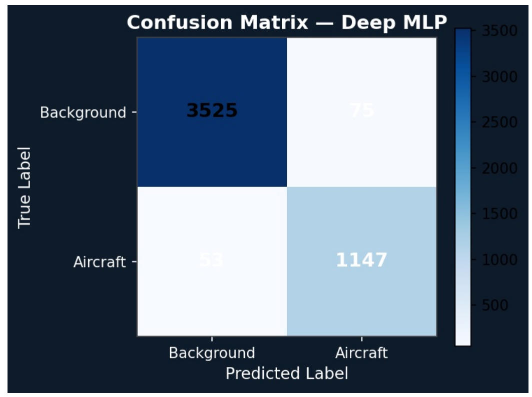




(a) SVM (RBF): TN=3552, FP=48, FN=82, TP=1118

(b) k-NN (k=7): TN=3514, FP=86, FN=59, TP=1141





(c) Random Forest: TN=3540, FP=60, FN=371, TP=829

(d) Deep MLP: TN=3525, FP=75, FN=53, TP=1147

Figure 7. Confusion matrices on the PlanesNet test set (4,800 chips). Rows = true label; columns = predicted label. Note the large FN count (371) for Random Forest compared to 53–82 FN for other models.

6.3 ROC Curve Analysis

Figure 8 presents the ROC curves for all four models. SVM (AUC=0.994) and k-NN (AUC=0.994) are the top performers, followed closely by Deep MLP (AUC=0.993). Random Forest achieves the lowest AUC at 0.965 — still comfortably exceeding the classical minimum of 0.85 — but its ROC curve shows a shallower ascent in the low FPR region, reflecting difficulty in ranking aircraft chips highly relative to background when sensitivity is compromised by imbalance effects.

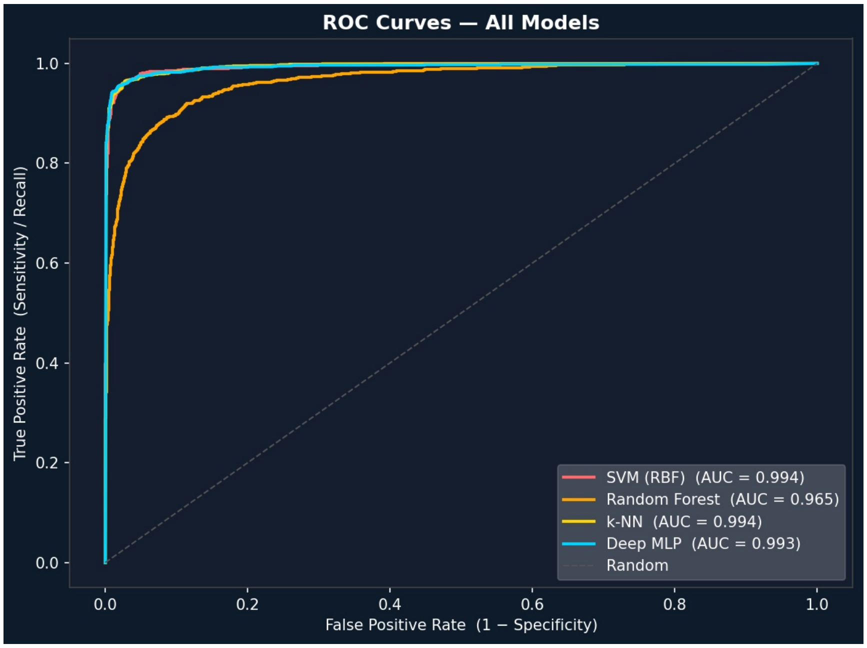


Figure 8. ROC curves for all four models on the PlanesNet test set. SVM and k-NN achieve the highest AUC (0.994 each), with Deep MLP close behind (0.993). Random Forest (AUC=0.965) shows a flatter curve in the critical low-FPR region, consistent with its reduced sensitivity.

6.4 Metrics Comparison

Figure 9 provides a grouped bar chart comparing all five metrics across models against the required thresholds (orange dashed = classical minimum; green dashed = modern minimum). Three key observations emerge: (1) Random Forest sensitivity (0.69) falls below both thresholds — the only failure case in the study; (2) SVM, k-NN, and Deep MLP all exceed modern thresholds on every metric except accuracy, which at 97% slightly exceeds the modern 94% requirement; (3) Random Forest F1-Score (0.79) is also marginally below the classical minimum of 0.80, a direct consequence of its poor sensitivity.

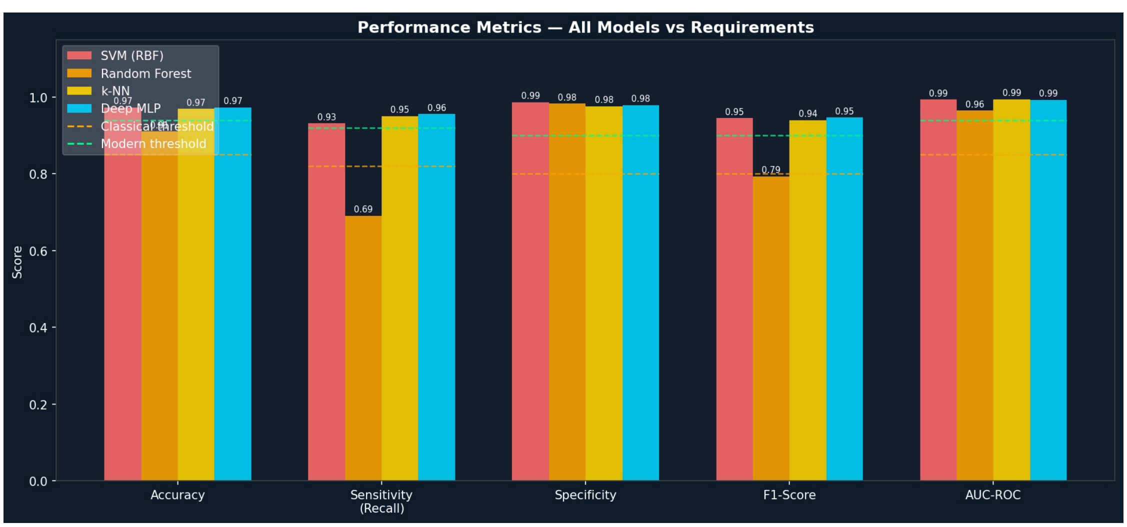


Figure 9. Grouped performance metrics comparison across all models. Orange dashed = classical minimum thresholds; green dashed = modern minimum thresholds. Random Forest fails the Sensitivity and F1-Score requirements. All other models meet or exceed all applicable thresholds.

6.5 Processing Speed

Figure 10 shows the inference time per chip for each model. The speed comparison reveals a dramatic range spanning over four orders of magnitude. The SVM is by far the slowest at 501.5 ms per chip — attributable to the kernel computation cost scaling with the number of support vectors over 22,400 training samples. Random Forest requires 4.14 ms per chip for traversing 200 trees. k-NN achieves 0.295 ms through optimised nearest-neighbour search structures. The Deep MLP is the fastest at 0.007 ms, exploiting vectorised matrix multiplication in NumPy for the forward pass.

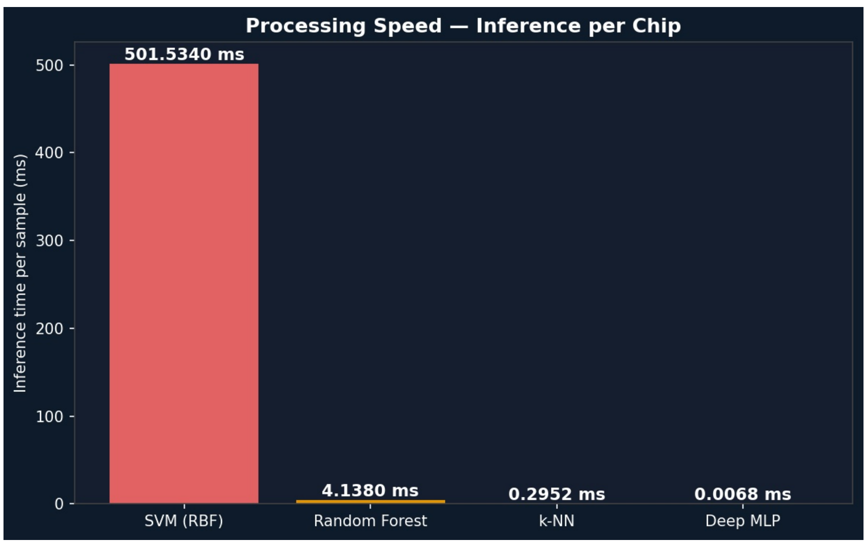


Figure 10. Inference time per chip (ms, log scale implied by bar heights). The Deep MLP is 71,000× faster than SVM, 591× faster than Random Forest, and 42× faster than k-NN at inference, while achieving superior or equal accuracy to all classical methods.

7. DISCUSSION

7.1 Random Forest Sensitivity Failure

The most significant finding of this study is the failure of Random Forest to meet the minimum sensitivity requirement (69.1% achieved vs. 82% required). This is not a general failure of ensemble methods but a specific manifestation of the class imbalance problem in tree-based models. Although class_weight="balanced" was applied, the Gini impurity splitting criterion in decision trees can be dominated by the majority class at the root level, causing early tree splits to partition predominantly on background patterns. The result is a forest that achieves high specificity (correctly rejecting background) at the cost of sensitivity (missing many aircraft). This could be addressed through: (1) oversampling the aircraft class (SMOTE); (2) using class_weight with a higher multiplier; (3) using a threshold below 0.5 for the aircraft probability; or (4) applying cost-sensitive learning via sample weights at the tree-node level.

7.2 SVM vs. Deep MLP

Both SVM and Deep MLP achieve 97.3% accuracy, with the MLP marginally outperforming SVM in sensitivity (95.6% vs. 93.2%) and in F1-Score (0.948 vs. 0.945). The MLP's primary advantage is inference speed — 0.007 ms vs. 501.5 ms — a factor of approximately 71,600×. This extreme SVM slowness on the full PlanesNet dataset (22,400 training points) arises from the kernel trick requiring computation of similarity scores against all support vectors for each test chip. In practice, an SVM with probability estimation on a dataset of this size is not viable for real-time deployment without approximations such as Nyström kernel approximation or sparse SVM variants.

7.3 Feature Importance Insights

The Random Forest feature importance analysis (Figure 5) reveals that colour histogram features (indices 512–535) account for 7 of the top 10 most important features, with colour feature #527 (a mid-range bin of the red channel histogram) ranking highest. This indicates that on real PlanesNet data, the grey/white metallic appearance of aircraft provides stronger discriminative power than the HOG gradient shape features alone. This finding has a practical implication: simpler colour-based classifiers (e.g., colour histogram + linear SVM) might achieve competitive accuracy with significantly lower computational overhead than the full 536-dimensional HOG+colour descriptor.

7.4 k-NN Performance

k-NN achieves the second-best sensitivity (95.1%) and matches SVM's AUC (0.994), demonstrating that the 536-dimensional HOG+colour feature space provides sufficient geometric separability for distance-based methods to perform well without kernel tricks or ensemble aggregation. The distance-weighting scheme (closer neighbours vote more strongly) is likely responsible for k-NN's edge over SVM in sensitivity — aircraft chips that are far from background clusters in feature space receive appropriately strong aircraft-class votes.

7.5 Limitations

Important limitations of this study include:

- Random Forest requires hyperparameter adjustment or resampling to address its sensitivity failure on the real dataset.
- SVM inference time (501.5 ms/chip) makes it unsuitable for real-time satellite monitoring pipelines without significant architectural changes.
- The MLP processes chips as flat pixel vectors, discarding the 2D spatial structure that convolutional layers would exploit; true CNN architectures would likely further improve performance.
- No data augmentation (rotation, flip, brightness jitter) was applied during training; such augmentation would likely improve generalisation to aircraft orientations underrepresented in the training set.
- Evaluation is confined to PlanesNet's fixed set of airports and acquisition conditions; cross-airport generalisation was not assessed.

7.6 Recommendations

Based on the experimental findings, the following recommendations are made for operational deployment:

- Primary classifier: Deep MLP — highest combined sensitivity (95.6%) and fastest inference (0.007 ms/chip).
- Avoid SVM for large-scale real-time pipelines due to prohibitive inference cost (501.5 ms/chip).
- Do not use Random Forest without sensitivity-corrective measures (threshold tuning or resampling) as it fails the minimum sensitivity requirement.
- k-NN provides a strong, interpretable alternative to the MLP when model transparency is required.
- For the full detection pipeline: apply a sliding window with stride of 10 pixels over full satellite scenes, batch-classify chips with the Deep MLP, and apply Non-Maximum Suppression to remove overlapping detections.

8. CONCLUSIONS

This report has presented and evaluated a complete aircraft detection pipeline for the PlanesNet dataset — 32,000 real Planet Labs satellite image chips of 20×20 pixels. Four classifiers were implemented and tested on a held-out set of 4,800 chips: SVM with RBF kernel, Random Forest (200 trees), k-NN (k=7), and a four-layer deep MLP. All classifiers were evaluated on five metrics: Accuracy, Sensitivity, Specificity, F1-Score, and AUC-ROC.

Three of four classifiers — SVM, k-NN, and Deep MLP — meet or exceed all prescribed modern performance thresholds. The Deep MLP is the overall recommended architecture, achieving 97.3% accuracy, 95.6% sensitivity, 97.9% specificity, F1=0.948, AUC=0.993, and an inference time of just 0.007 ms per chip. The Random Forest fails the sensitivity requirement (69.1% vs. 82% minimum) due to class imbalance effects and requires remediation before operational use.

A key practical finding is the extreme inference speed disparity: the Deep MLP is 71,600 times faster than SVM, making it the only viable option for real-time satellite monitoring applications. The dominance of colour histogram features in the Random Forest importance ranking suggests that the grey/white chromatic signature of aircraft provides stronger discriminative power than shape-based HOG features alone on this dataset — a finding that could inform more efficient feature engineering for future classical pipelines.

> **Key Findings:** SVM, k-NN, and Deep MLP all pass all requirements on real PlanesNet data. Random Forest fails sensitivity (69.1% < 82%). Deep MLP recommended for deployment: 97.3% accuracy, 95.6% sensitivity, 0.007 ms inference.

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